Understanding Docker and Apache Airflow

Task. Help understand documentation of Docker and Apache Airflow.

Model. Claude Opus 5.5

Prompt

> I'm a student building a data warehouse project. I'm comfortable with data modelling (star schemas, fact and dimension tables, ETL concepts) but new to [Apache Airflow / running Airflow in Docker].
>
> Act as a teacher, not a coder: do not write code or config for me. Instead:
>
> 1. Explain what [the tool / the integration] is for and what problem it solves, in a few short paragraphs.
> 2. Map its core ideas to data modelling concepts I already know, and say where each analogy breaks down.
> 3. Point me to the parts of the official documentation I should read first, in order.
> 4. End with 3 questions I should be able to answer once I understand it, so I can check myself.
>
> If something depends on a version or setup detail, ask me instead of assuming.

LaTeX and ACM Template Formatting

Task. LaTeX formatting and ACM template formatting.

Model. Claude Opus 5.5

Prompt

Our report uses the ACM `acmart` class with the `sigconf` option (see `paper.tex`). Please reformat every table in `paper.tex` and `chapters/*.tex` so they follow the ACM template:

- Use `booktabs` rules (`\toprule`, `\midrule`, `\bottomrule`) and no vertical lines.
- Put the caption above the table, and give each table a `\label` and a `\Description` for accessibility.
- Make tables fit the column (`\columnwidth`, with `tabularx` for long text); use `table*` only when a table cannot fit in one column.
- Right-align numbers and use thousands separators.

Do not change any numbers or wording inside the tables. When you're done, list each table you changed and what you changed, and point out any table you weren't sure about.

Figure Creation Using TikZ

Task. Figure creation using TikZ.

Model. Claude Opus 5.5

Prompts

Schema

Using the warehouse schema in our DDL (fact tables `fact_service` and `fact_stop`, dimension tables `dim_date`, `dim_time`, `dim_station`), create a TikZ entity-relationship diagram for our ACM two-column paper.

- Make it a `standalone` file in `figures/`, so it can be compiled to PNG/PDF and included with `\includegraphics`.
- Show each table with its columns; mark primary keys and foreign keys.
- Visually separate fact tables from dimension tables, and draw one line per foreign key with 1 and N labels.
- Match the colours of our other figures (accent `#2F5D8C`) and keep text readable at full page width.

Take the table and column names from the DDL; don't make any up. Compile the file and check that nothing overlaps.

Query plans

For each query in `dutch_railway/qpo/queries/` (skip `_example`), create a TikZ figure for our ACM paper that compares the baseline plan with the optimized plan.

- Put the baseline on the left and the optimized plan on the right, as bottom-up operator trees (tables at the bottom, result at the top).
- Highlight the operators that differ between the two plans in one accent colour, and add a one-line note under each side saying why the optimized version is faster.
- Use real numbers from the saved `EXPLAIN ANALYZE` output in `qpo/results/` where they are available (row counts, loops). If a plan isn't saved or looks stale, draw that side from the SQL and tell me.
- Use one shared style file and match our existing figures (`figures/etl_dag.tex`). Make each figure a `standalone` file that compiles to PNG.
- Insert the figures into `chapters/qpo.tex` with captions and `\Descriptions`.

Compile every figure, check that the layouts don't overlap, and tell me about anything you could not verify.